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An Enhanced Gelled Acid System for Fracturing with Proppant to Increase Production in a Middle East Field

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Abstract

Proppant fracturing treatments in sandstone and shale formations are routinely executed. However, when carbonate formations are the target, acid fracturing is the preferred treatment method. Acid fracturing delivers a high initial production, however maintaining a sustainable production rate is a challenge in the tight cretaceous carbonate formations. These formations require highly conductive fractures to enhance production, the creation of long differently etched fractures is necessary. Certain challenges are encountered in acid fracturing including high leak-off and fast acid reaction rates. The conventional practice of acid fracturing involves complex pumping schemes of pad, acid and viscous diverter fluids to achieve fracture length and conductivity targets. A simplified and cost-effective approach to acid fracturing is designed to be deployed to wells in a gas producing formation in the Middle East. This enhanced gelled system is a single fluid system that delivers rock-breaking viscosity, slow-reactive acid, and effective leak-off control. Combining this system with proppants gets the benefits from both treatments so long as proppant embedment is taken into consideration.

This enhanced gelled system reduced three fluid requirements to one by eliminating the need for an intricate pumping schedule that otherwise would include: a non-acid fracturing pad stage to breakdown the formation and generate the targeted fracture geometry; a retarded emulsified acid system to achieve deep penetrating differently etched fractures, and a self-diverting agent to minimize fluid leak-off. The enhanced gelled system was prepared with 10 and 15% hydrochloric acid (HCl) and a polymer with delayed crosslinking using a zirconium metal in unspent acid. The fluid design optimization details are included to illustrate how a single system can replace the need for multiple fluids. Rheology testing at temperatures up to 285°F, proppant settling tests for ceramic (light-weight proppant) LWP and ceramic (high-strength proppant) HSP up to 5 ppa, and static breaking were conducted.

Results of the rheology and proppant settling tests with this new system exceeded expectations. Furthermore, the crosslinked system could aid acid leak-off and was delayed to reduce friction pressure so that optimal pump rates were achieved to improve fluid placement across the entire open-hole lateral stage. An enhanced gelled system is desirable to prevent acid from reacting in the near-wellbore and reaching as far out into the generated fracture as possible. Full acid reaction rate was not reached with this system until the release of the internal breaker.

This study demonstrates how common acid fracturing design and operational practices can be simplified by using a cost-effective approach using a single enhanced gelled acid system.

Introduction

In the field of oil and gas well stimulation, particularly in carbonate formations, acidizing has long been a vital technique for enhancing reservoir permeability and productivity. However, conventional acid systems often react too quickly with the rock matrix near the wellbore, resulting in limited acid penetration and suboptimal stimulation outcomes. Formation damage can also be inevitable in the absence of the right additives (Assem et al. 2013; Assem et al. 2019). To address these challenges, delayed acid systems have been developed and refined over the past two decades, offering improved control over acid-rock reactions, enhanced stimulation effectiveness, and better operational safety. These systems, also known as retarded or controlled-release acid systems, have seen substantial innovation in formulation, deployment, and performance evaluation since the early 2000s.

An important milestone in the evolution of delayed acid systems was the introduction of chelating agents and organic acids as alternatives to HCl. Organic acids such as acetic, formic, and citric acids have inherently slower reaction rates with carbonate formations, making them suitable candidates for delayed acid systems. Additionally, they pose less corrosion risk to tubulars and are safer to handle at the surface. Over the years, the blending of these organic acids with traditional systems has resulted in tailored acidizing fluids that maintain a balance between reaction control and dissolution efficiency (Buijse et al. 2004).

One of the significant advances in delayed acid technology has been the development of emulsified acid systems. These formulations, typically composed of hydrochloric acid (HCl) emulsified in oil with the help of surfactants, create a diffusion barrier that slows down the acid's contact with the carbonate surface. This mechanism allows deeper penetration before significant reaction occurs, minimizing near-wellbore damage and increasing the radius of stimulation. Research by Al-Mutairi et al. (2009) demonstrated the superior performance of emulsified acids in high-temperature carbonate formations, showing how these systems could retain up to 60% of the acid in the emulsion after several hours, thereby extending the effective reaction time window. Building on this foundation, Sayed et al. (2014) explored the performance of emulsified acids in oil-saturated carbonate cores. Their experimental studies revealed that crude oil saturation significantly increases the acid volume required to achieve breakthrough compared to water-saturated conditions. Despite the additional acid requirement, the emulsified system retained its delayed reactivity, allowing deeper wormhole propagation and improved stimulation coverage in oil-wet environments.

In parallel, the oil industry has seen the rise of polymer-based delayed acid systems, particularly gelled acid and viscoelastic surfactant (VES)-based fluids. These systems use polymers or surfactants to increase the viscosity of the acid solution, creating a transport medium that reduces acid diffusion rates. This controlled reactivity not only helps in minimizing face dissolution near the wellbore but also enhances wormhole propagation deeper into the formation. The use of VES-based acids, in particular, gained momentum after the mid-2000s due to their self-breaking nature and ease of cleanup. Al-Muntasheri et al. (2009) noted the benefits of these systems in achieving uniform acid distribution and reducing fluid loss in heterogeneous carbonate reservoirs. The viscoelastic properties help in suspending solid particles and delivering the acid more uniformly across the reservoir face, a major challenge in conventional acidizing treatments.

Temperature resistance has been a persistent challenge in designing delayed acid systems, especially for deep or high-pressure/high-temperature (HPHT) wells. In recent years, advances in high-temperature-resistant surfactants and stabilizers have expanded the applicability of VES-based and polymer-acid systems to environments previously deemed too hostile for acidizing. According to a study by Wang et al. (2012), modified viscoelastic surfactants showed stable performance at temperatures exceeding 300°F, maintaining their retarding properties without premature degradation. These systems were further optimized

by integrating thermal stabilizers and pH buffers to ensure consistent performance throughout the treatment cycle.

A notable evolution within this domain is the integration of gelled acid fracturing with proppant placement, which addresses two key challenges: the rapid acid reaction rate and the long-term conductivity of created fractures (Liu et al. 2024). Traditional acid fracturing may create conductivity via etched channels, but these can close under formation stress. By combining gelled acid with solid proppants—such as resin-coated sand or ceramic materials—operators can maintain fracture conductivity over time. This hybrid approach is particularly beneficial in deep carbonate reservoirs where maintaining open fracture pathways is crucial for sustained production. Bale et al. (2010) summarized that gelled acid carrying proppant can create acid-etched channels while simultaneously placing proppant in strategic zones, enhancing both near- and far-field conductivity. The gel helps suspend the proppant during pumping and slows acid reaction, thereby increasing contact time and effectiveness. Assem and Nasr-El-Din 2017 and Assem et al. (2017) further investigated how different acid systems interact with sand and ceramic proppants. Their findings revealed that some acid formulations—especially mud acid—could degrade proppant materials, reducing long-term fracture conductivity. These insights emphasized the importance of acid-proppant compatibility when designing effective fracturing treatments in carbonate environments.

Nanotechnology has also begun to influence the design and application of delayed acid systems. Researchers have investigated the use of nanoparticles to control acid release rates and enhance acid-rock interaction selectivity. Nanoparticle-based in situ gelled acid systems have shown superior efficiency in diverting acid within carbonate reservoirs under harsh conditions by forming multiple wormholes and reducing acid leak-off through nanoparticle aggregation in small pore throats. Experimental and modeling results revealed that factors like salt concentration, pH, and shear rate significantly influence gelation kinetics and viscosity, directly affecting the diversion performance (Abdelfatah et al. 2017).

In the past 20 years, significant advancements in delayed acid system technologies have been witnessed. Innovations in fluid chemistry—ranging from emulsified acids and viscoelastic surfactants to nanotechnology, hybrid acid-fracturing techniques, and organic acid blends—have extended the utility of acidizing treatments to increasingly challenging reservoir conditions. Improvements in simulation models and real-time diagnostics have further enhanced the design and execution of acidizing operations. As the industry continues to prioritize sustainability and performance, the development of smart, adaptive, and environmentally responsible acid systems is likely to shape the next generation of well stimulation technologies.

The objectives of this laboratory study were to: evaluate the enhanced gelled system and its ability to carry proppants up to 5 ppa.

Favorable results meant approval for injection in a Middle East field.

Experimental Studies

Proppant fracturing treatments in sandstone and shale formations are routinely executed. However, when carbonate formations are the target, acid fracturing is the preferred treatment method. Acid fracturing delivers a high initial production however, maintaining a sustainable production rate is a challenge in the tight cretaceous carbonate formations. These formations require highly conductive fractures to enhance production, the creation of long differently etched fractures is necessary. Certain challenges are encountered in acid fracturing.

Rheology

Enhanced gelled system blend with 10 and 15% HCl acid was blended at the laboratory. Several acid formulations were well mixed, where the components were added in the following order: corrosion inhibitor, acid gellant, suspension additive. After mixing for 5 minutes, non-emulsifier and iron control agent were

added. Linear gel was allowed at least one hour to hydrate. Finally, the "on the fly" additives like crosslinker and breaker (if needed) were added to prepare crosslinked gel for rheology study.

Rheology studies were conducted on a Chandler HTHP 5550 viscometer at test temperatures of 285°F and 270°F. Tests were performed with 50 mL of the acid system and using a B5 bob and R1 sample cup. Fluids utilized in the optimization studies were prepared by this procedure.

Proppant Settling

The procedure is very similar to rheology testing. Right after the crosslinker was added to prepare crosslinked gel, up to 5 ppa of the 40/70 ceramic (light-weight proppant) LWP or 40/80 ceramic (high-strength proppant) HSP were added and thoroughly mixed. The gel was then transferred to a test tube through a funnel and monitored for at least 1 hour at ambient temperature.

Static Breaker

The same procedure for rheology testing was implemented. Optimal formulations were used to prepare 10 and 15% HCl enhanced gelled system, respectively. The breaker loading varied between 0 and 8 ppt. The crosslinked gel samples were transferred to test tubes and placed in the oven at 250°F for up to 48 hours and the gel strength was checked at various times.

Spent Acid Testing

Rheology was optimized for the 15% HCl enhanced gelled system at 285°F and 270°F and for 10% HCl acid system at both temperatures. As the concern of acid spending was raised, a general formulation that worked at different concentrations was sought by eliminating the effect of the crosslinker loading.

Results and Discussion

Rheology

Rheology was optimized for 285°F applications. The rheology for optimal formulation is shown in [Fig. 1](#). Desired rheology results were observed in the absence of the breaker. The fluid showed viscosity exceeding the 300-cP minimum requirement during the 1-hour period. This fluid was successfully optimized and fell within the required viscosity specifications.

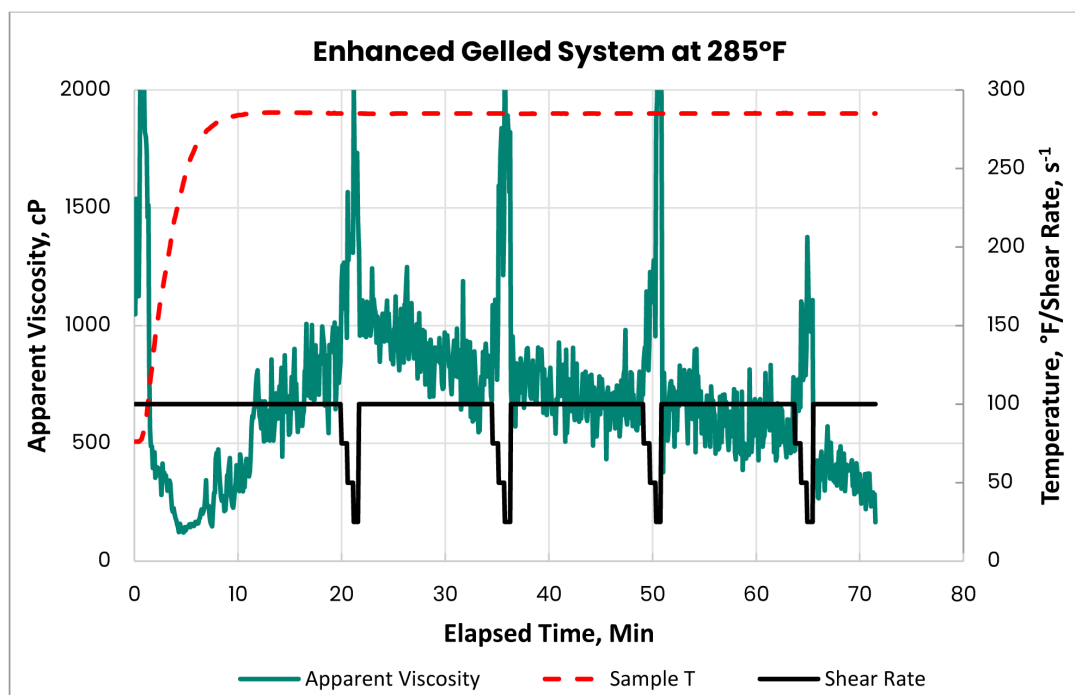


Figure 1—15% HCl Enhanced Gelled System Rheology at 285°F without Breaker.

Breaker loading was tested at 3 and 7 ppt (Fig. 2). The rheology profile in the presence of the breaker showed a rather rapid decrease in viscosity in rheometer testing. However, after the rheology testing, the majority of the gel inside the testing tube showed strong crosslinked structure. Only a portion of gel inside the testing tube was subject to shear and eventually broke. Since, dynamic breaker testing did not generate uniform breaking profile, static breaker testing was performed.

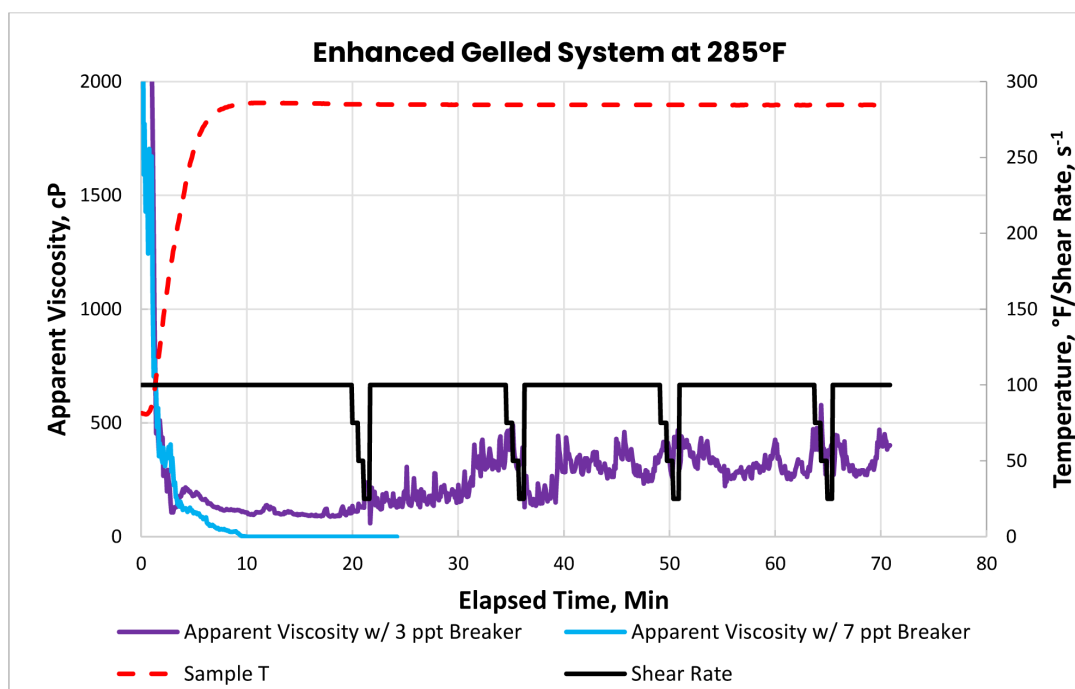


Figure 2—15% HCl Enhanced Gelled System Rheology at 285°F with 4 and 7 ppt of Breaker.

The tests were repeated for 10% and the optimal formulation is shown in Fig. 3. Desired rheology results were observed in the absence of the breaker. This fluid passed the 300-cP minimum viscosity requirement. Breaker loading was evaluated at 3, 4 and 5 ppt (Fig. 4). Similar to the results from 15% testing, there was a general rapid decrease in viscosity while 3 ppt of breaker seemed insufficient. As mentioned previously, those results are not conclusive due to most of the fluid still being crosslinked after the test was performed. Static break is the preferred method to determine breaker loading requirements.

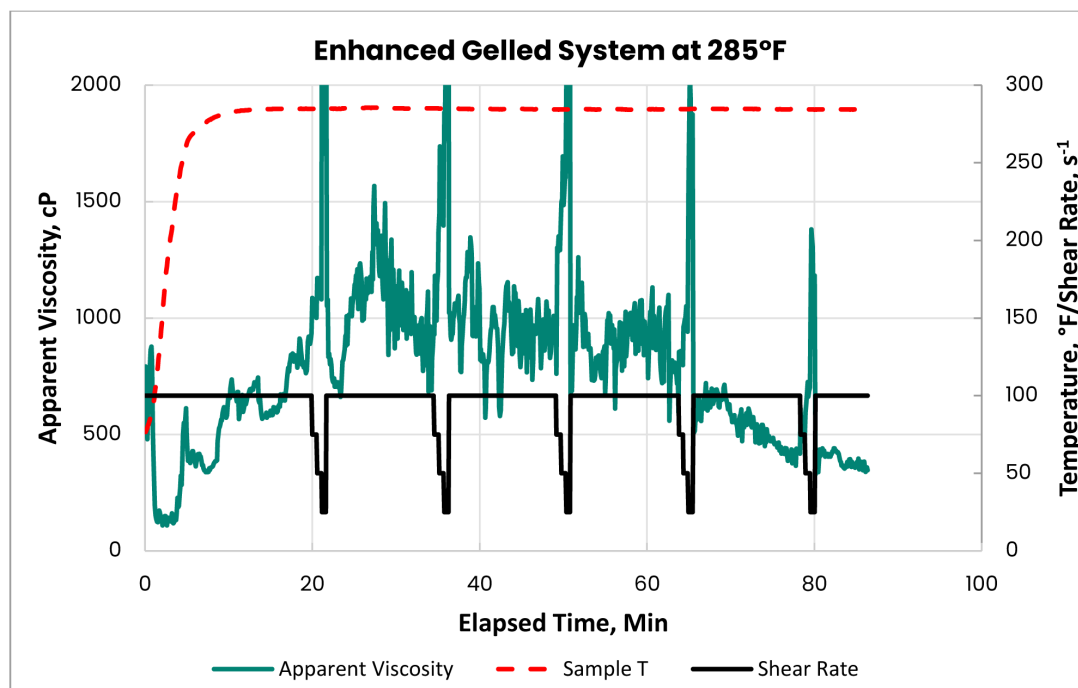


Figure 3—10% HCl Enhanced Gelled System Rheology at 285°F without Breaker.

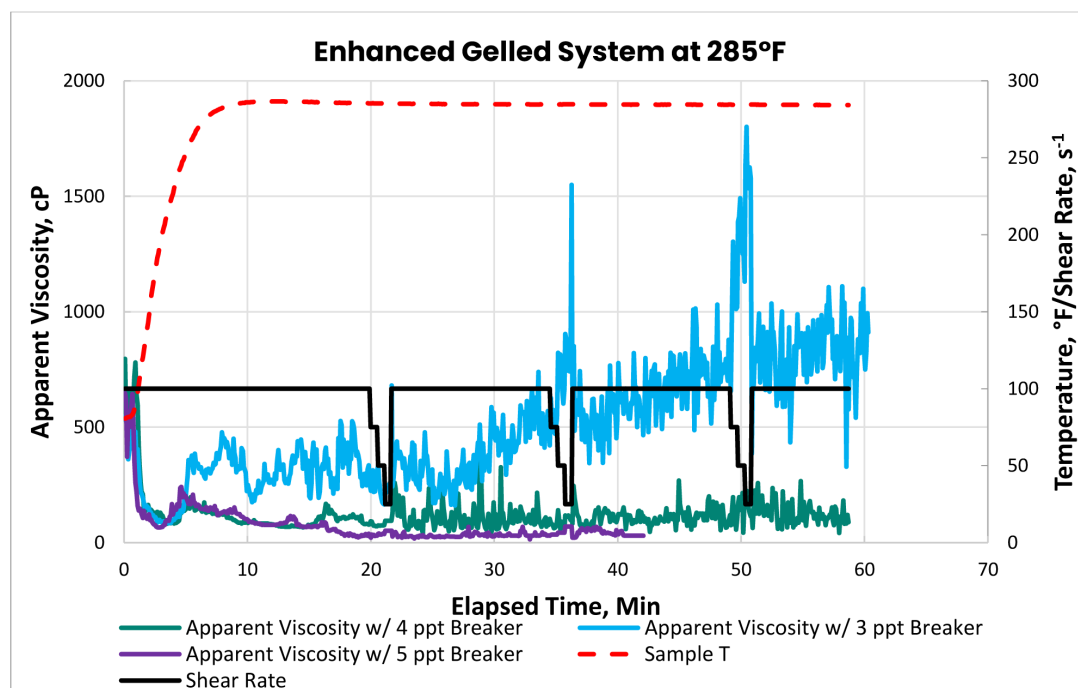


Figure 4—10% HCl Enhanced Gelled System Rheology at 285°F with 3, 4, and 5 ppt of Breaker.

Similar optimization was conducted at 270°F. The rheology of optimal formulations is shown in Figs. 5 and 6, respectively. Desired rheology results were observed in the absence of the breaker. All the fluids passed the minimum viscosity requirement of 300 cP. Breaker loading was evaluated at 4 ppt for both formulas, respectively. Both showed a rapid decrease of viscosity in rheometer testing. The focus was then shifted to static breaking testing.

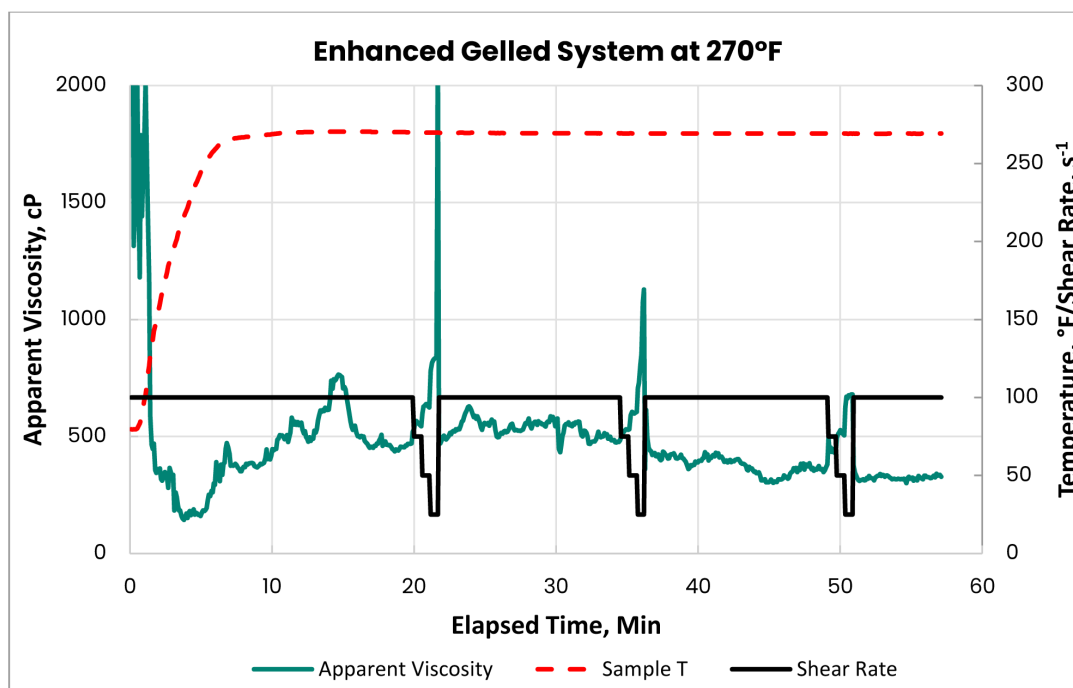


Figure 5—15% HCl Enhanced Gelled System Rheology at 270°F without Breaker.

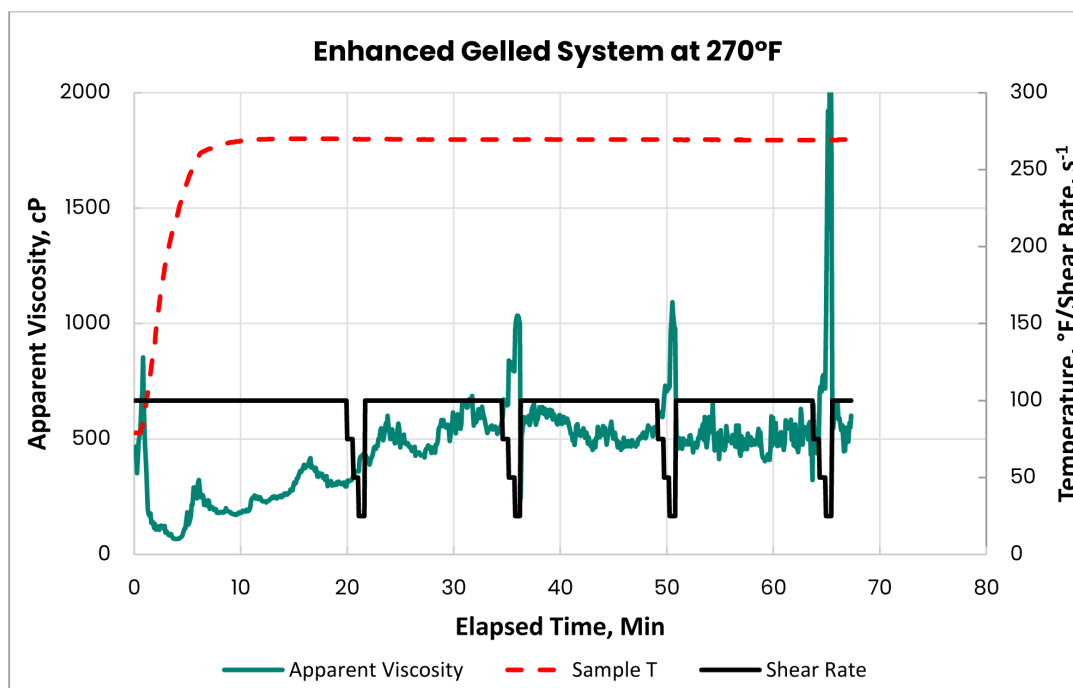


Figure 6—10% HCl Enhanced Gelled System Rheology at 270°F without Breaker.

Proppant Settling

Figs. 7 and 8 show the results of 40/70 ceramic (light-weight proppant) LWP and 40/80 ceramic (high-strength proppant) HSP with the 15% HCl enhanced gelled system at 4 and 5 ppa proppant loading, respectively. After an hour, good suspension of the proppant was observed ($> 90\%$ remained suspended). Figs. 9 and 10 show the results of these proppants with the 10% acid system at 5 ppa proppant loading. In the case of a lower crosslinker loading, linear gel was preheated to 175°F before adding the crosslinker and proppant.

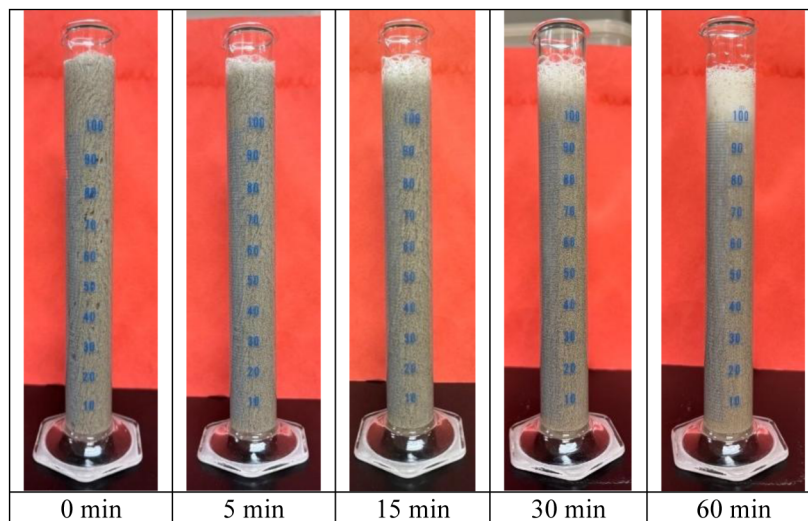


Figure 7—4 ppa of 40/70 ceramic LWP Proppant Settling Test in 15% HCl Enhanced Gelled System.

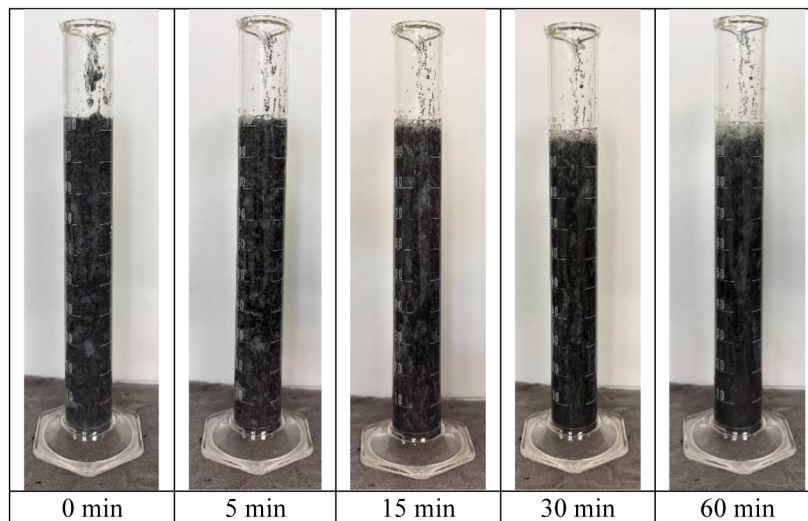


Figure 8—5 ppa of 40/80 ceramic HSP Proppant Settling Test in 15% HCl Enhanced Gelled System.

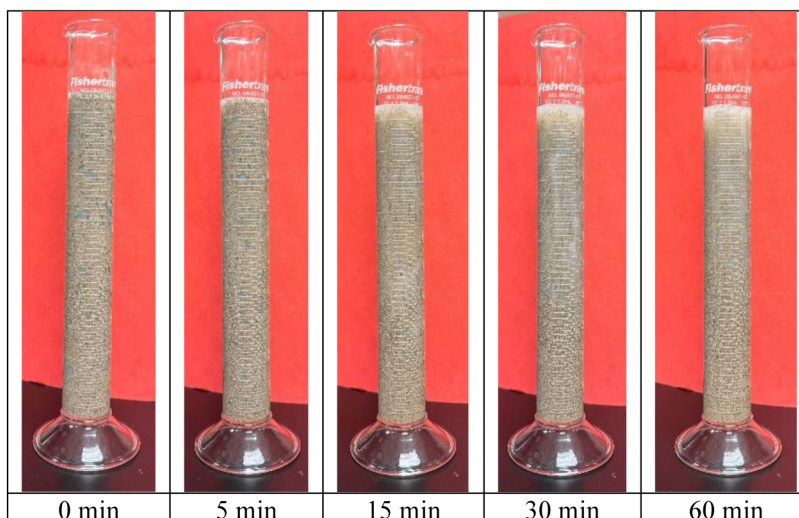


Figure 9—5 ppa of 40/70 ceramic LWP Proppant Settling Test in 10% HCl Enhanced Gelled System.

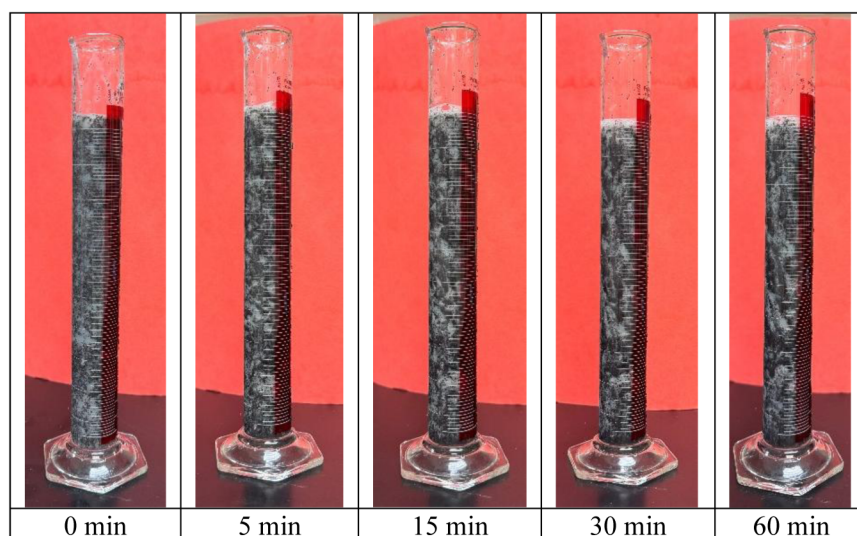


Figure 10—5 ppa of 40/80 ceramic HSP Proppant Settling Test in 10% HCl Enhanced Gelled System.

Static Breaker

Static breaker tests were performed at 250°F. When the breaker was absent, the crosslinked polymer did not completely lose viscosity (stringy) even after 48 hours of testing. However, it took approximately 24 to 48 hours to have the crosslinked polymer totally lose viscosity. When the breaker loading was increased above 5 ppt, the crosslinked gel would break closer to 24 hours. Fig. 11 shows a visual comparison between a crosslinked sample versus a completely broken (water-like) sample.

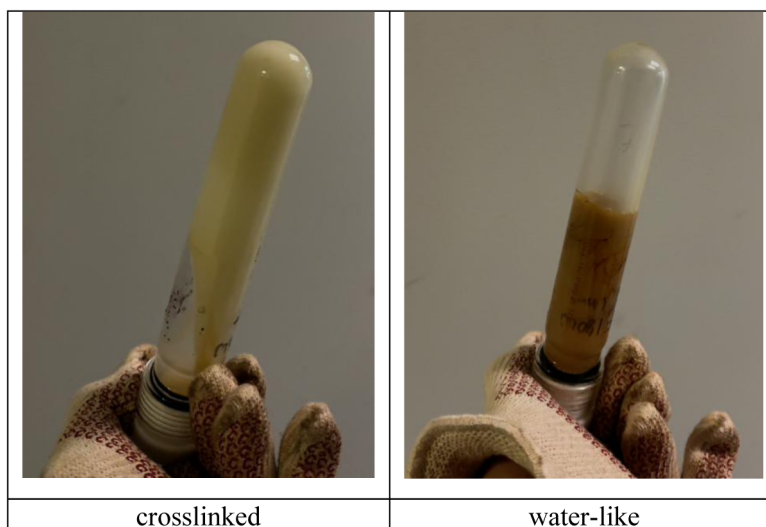


Figure 11—An example of crosslinked sample versus water like sample from static breaker testing.

Spent Acid Testing

Rheology testing was performed on the assumption that a 15% HCl crosslinked fluid would be spent to 10% HCl during a 1-hour pumping, while a 10% HCl would be spent to 7%. Based on the first optimization, the 15% HCl enhanced gelled system used different loading of crosslinker compared to 10%.

In order not to have variations on the compositions of the fluids more optimization was performed and the same crosslinker loading achieved the desired rheology (above 300 cP) for 15%, 10% and 7% enhanced gelled system, respectively. The rheology of the new optimized formulations is shown in Figs. 12 and 13.

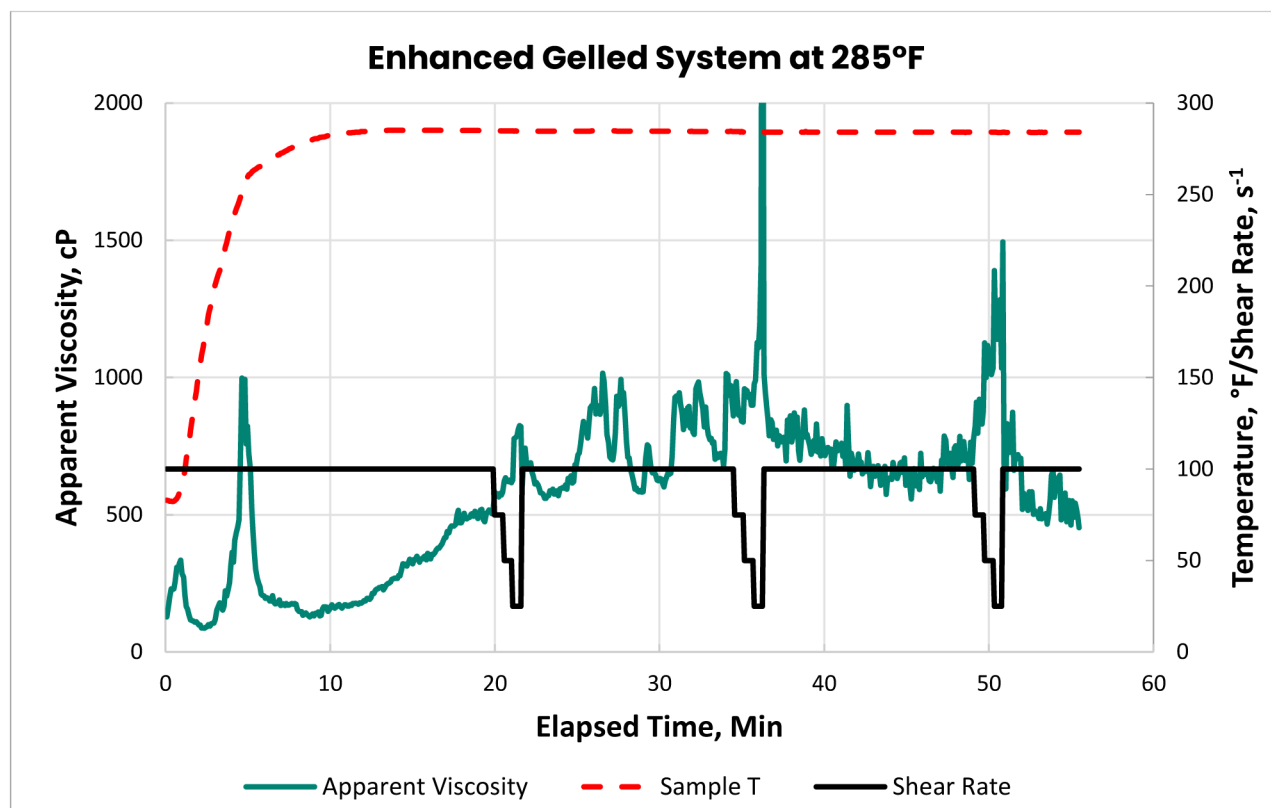


Figure 12—7% Enhanced Gelled System Rheology at 285°F without Breaker.

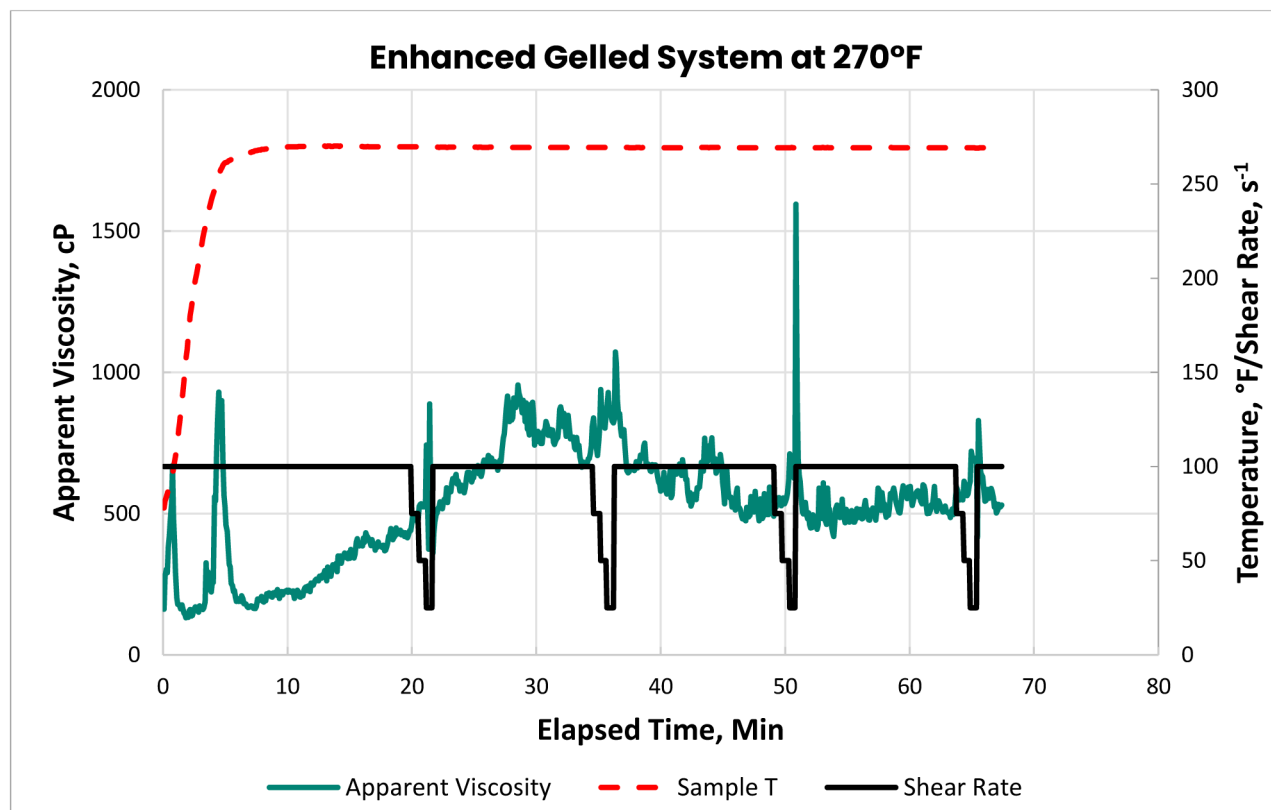


Figure 13—7% Enhanced Gelled System Rheology at 270°F without Breaker.

Figs. 14 and 15 show the results of 40/70 ceramic LWP and 40/80 ceramic HSP proppant settling test with the 10% HCl enhanced gelled system at 5 ppa proppant loading, respectively. Desert Pink Limestone rock pieces were placed in both graduated cylinders. After an hour, good suspension of both proppants was observed (similar to Figs. 9 and 10).

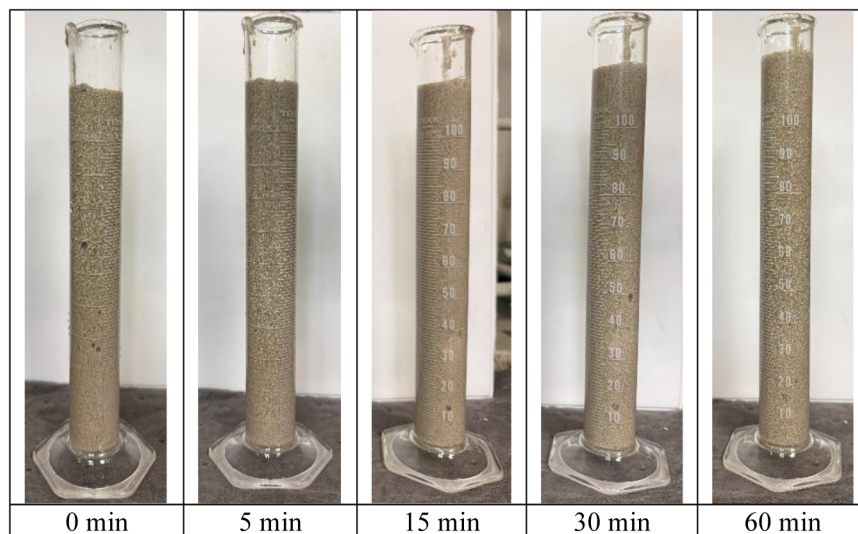


Figure 14—5 ppa of 40/70 ceramic LWP Proppant Settling Test in 10% HCl Enhanced Gelled System with Desert Pink Limestone.

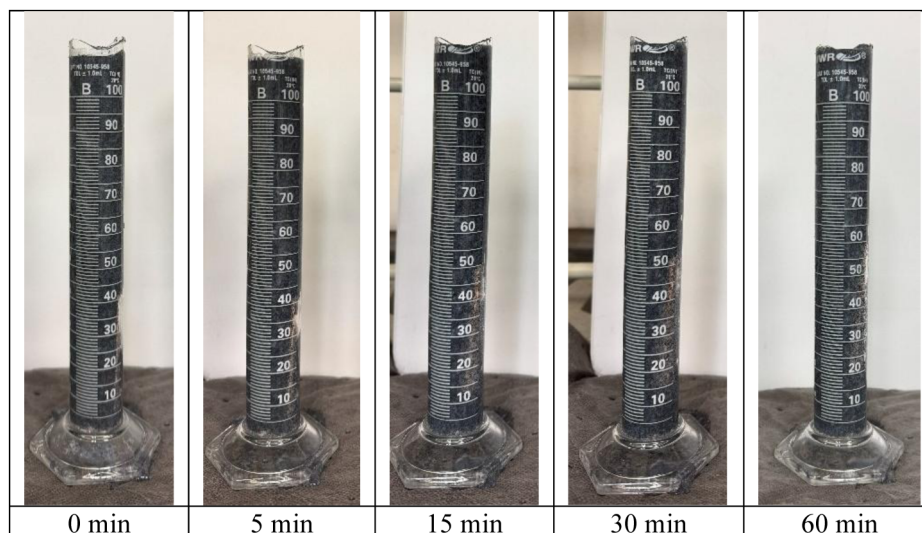


Figure 15—5 ppa of 40/80 ceramic HSP Proppant Settling Test in 10% HCl Enhanced Gelled System with Desert Pink Limestone.

Conclusions

Based on the results of the fluid optimization studies for 10 and 15% HCl enhanced gelled system, several observations can be summarized as the following:

- An optimal formulation was achieved that works at 285 and 270°F.
- The average viscosity was maintained above 300 cP for all optimized formulations for at least 1 hour.
- Both 15 and 10% HCl enhanced gelled system were able to suspend both 5 ppa of both, 40/70 ceramic LWP and 40/80 ceramic HSP, for at least 1 hour.
- Static breaker testing shows that the gel completely breaks between 24 and 48 hrs.
- Lower loading of breaker should be pumped in the beginning of the job and a higher loading can be pumped toward the end of the job.

Based on these results the enhanced gelled system could be used as an acid fracturing fluid candidate for proppant fracturing of carbonate formations.

References

- Abdelfatah, E., Bang, S., Pournik, M. et al 2017. Acid Diversion in Carbonates with Nanoparticles-Based in Situ Gelled Acid. Presented at the Abu Dhabi International Petroleum Exhibition & Conference, Abu Dhabi, UAE, 11–15 November. SPE- 188188-MS. <https://doi.org/10.2118/188188-MS>.
- Al-Muntasheri, G. A., Nasr-El-Din, H. A., Al-Noaimi, K. R. et al 2009. A Study of Polyacrylamide-Based Gels Crosslinked with Polyethyleneimine. *SPE J.* **14** (2): 245–251. SPE-105925-PA. <https://doi.org/10.2118/105925-PA>.
- Al-Mutairi, S. H., Nasr-El-Din, H. A., Hill, A. D. et al 2009. Effect of Droplet Size on the Reaction Kinetics of Emulsified Acid with Calcite. *SPE J.* **14** (4): 606–616. SPE-112454-PA. <https://doi.org/10.2118/112454-PA>.
- Assem, A. I., Kumar, H. T., Nasr-El-Din, H. A. et al 2019. Location and Magnitude of Formation Damage Due to Iron Precipitation During Acidizing Carbonate Rocks. *J Petrol Sci Eng.* **179**: 337–354. <https://doi.org/10.1016/j.petrol.2019.04.073>.
- Assem, A. I. and Nasr-El-Din, H. A. 2017. Interactions between Different Acids and Bauxitic-Based Ceramic Proppants Used in Gravel-Packed and Fractured Wells. *J Petrol Sci Eng.* **158**: 441–453. <https://doi.org/10.1016/j.petrol.2017.08.075>.
- Assem, A. I., Nasr-El-Din, H. A., and De Wolf, C. A. 2013. Formation Damage Due to Iron Precipitation in Carbonate Rocks. Presented at the SPE European Formation Damage Conference and Exhibition, Noordwijk, the Netherlands, 5–7 June. SPE- 165203-MS. <http://dx.doi.org/10.2118/165203-MS>.

- Assem, A. I., Nasr-El-Din, H. A., Fuss, T. et al 2017. Mud Acid Interactions with Sand and Clay-based Ceramic Proppants Used in Gravel-Packed and Fractured Wells. *SPE Prod & Oper* **32** (2): 196–207. SPE-175833-PA. <http://doi.org/10.2118/175833-PA>.
- Bale, A., Bang, S., and Klein, H. H. 2010. Stimulation of Carbonates Combining Acid Fracturing With Proppant (CAPF): A Revolutionary Approach for Enhancement of Sustained Fracture Conductivity and Effective Fracture Half-Length. Presented at the SPE Annual Technical Conference and Exhibition, Florence, Italy, 19–22 September. SPE-134307-MS. <https://doi.org/10.2118/134307-MS>.
- Buijse, M., De Boer, P., Breukel, B. et al 2004. Organic Acids in Carbonate Acidizing. *SPE Prod & Fac* **19** (3): 128–134. SPE- 82211-PA. <https://doi.org/10.2118/82211-PA>.
- Liu, P., Hu, H., Chen X. et al 2024. The Influencing Parameters and Methods to Enhance Acid-Etched Fracture Conductivity: A Review. *Geoenergy Sci. Eng.* **238**: 212844. <https://doi.org/10.1016/j.geoen.2024.212844>.
- Sayed, M. A., Assem, A. I., and Nasr-El-Din, H. A. 2014. Effect of Oil Saturation on the Flow of Emulsified Acids in Carbonate Rocks. *SPE Prod & Oper* **29** (1): 29–41. SPE-152844-PA. <http://dx.doi.org/10.2118/152844-PA>.
- Wang, G., Nasr-El-Din, H. A., Zhou, J. et al 2012. A New Viscoelastic Surfactant for High Temperature Carbonate Acidizing. Presented at the SPE Saudi Arabia Section Technical Symposium and Exhibition, Al-Khobar, Saudi Arabia, 8–11 April. SPE- 160884-MS. <https://doi.org/10.2118/160884-MS>.